Practical - 10

Practical Title: Secure Login/Logout with Sessions, Role-Based Access, Timeout, and Remember-Me Token.

Develop secure login and logout functionality using PHP sessions. Verify password using password_verify(), regenerate session ID after login, restrict dashboard access, implement role-based redirection, session timeout, and optional secure remember-me token.

A. Theory

1. **Are sessions started and destroyed securely?**

Yes. Sessions are started using session_start() and destroyed during logout using session_destroy(). Session cookies are configured with security settings such as HttpOnly and SameSite.

2. **Is password_verify() used correctly?**

Yes. The password_verify() function compares the entered password with the hashed password stored in the database to authenticate the user securely.

3. **Is unauthorized dashboard access blocked?**

Yes. Protected dashboard pages check whether a valid session exists. If the user is not logged in or the session has expired, access is denied and the user is redirected to the login page.

4. **Is session timeout implemented?**

Yes. A session timeout is implemented to expire inactive sessions after 15 minutes (900 seconds), requiring the user to log in again.

5. **Are admin and student users redirected to appropriate dashboards?**

Yes. After successful login, the user's role is checked. Admin users are redirected to the admin dashboard, while student users are redirected to the student dashboard.

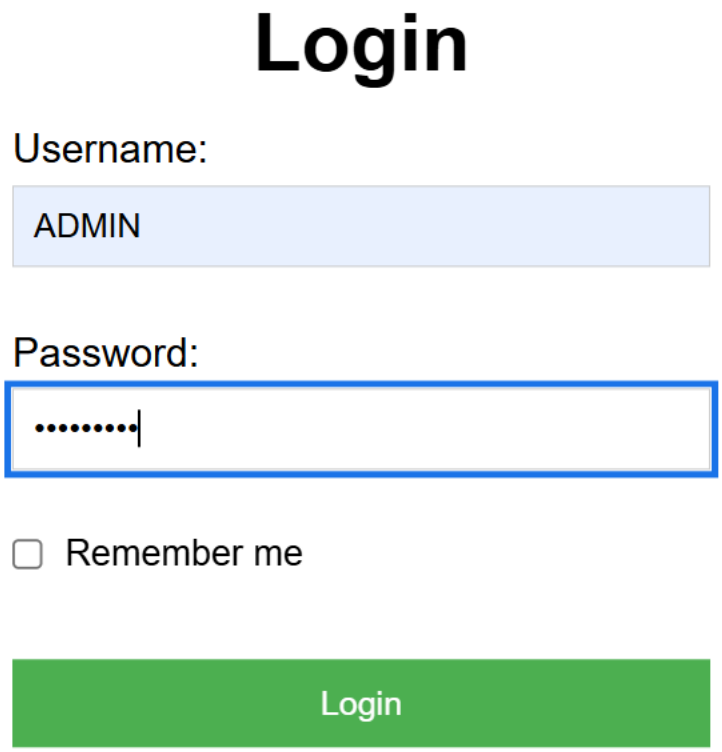
Implementation/ Output/Result

Tags:

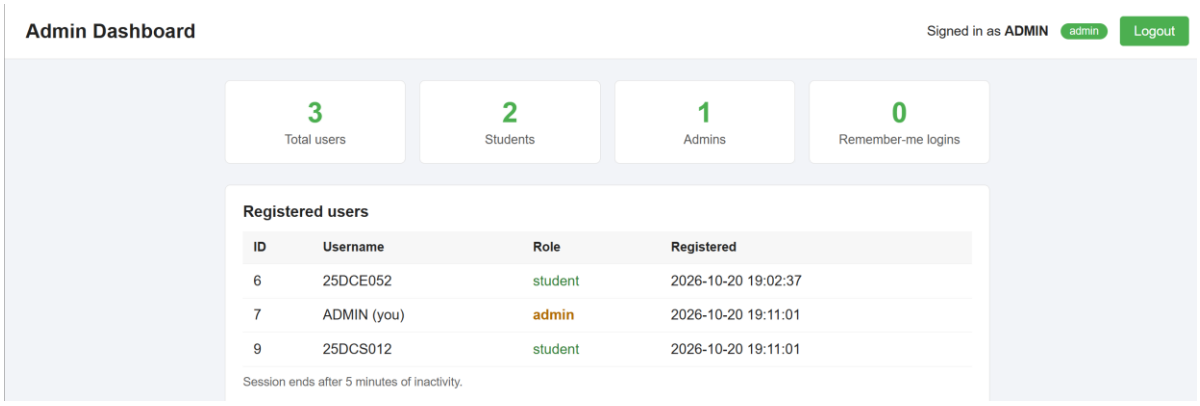
- Login form validation using JavaScript and PHP.
- User authentication using MySQL database credentials.
- Secure password verification using password_verify().
- Session management using PHP sessions.
- Session ID regeneration after successful login.
- Automatic session timeout after 5 minutes of inactivity.
- Role-based access for students and administrators.
- Secure logout by destroying the session.
- Redirecting users to their respective dashboards after login.

Output Screenshots:

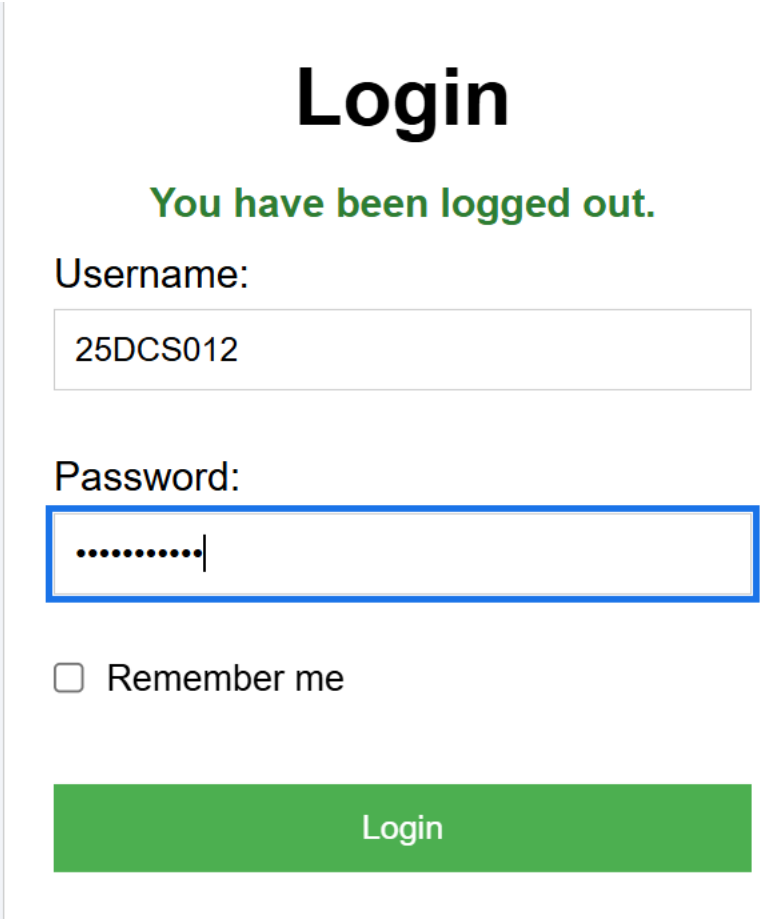
Admin Login Credentials:



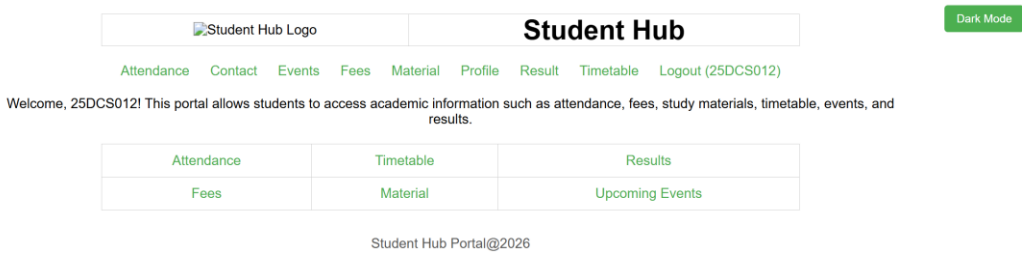
Admin Dashboard:



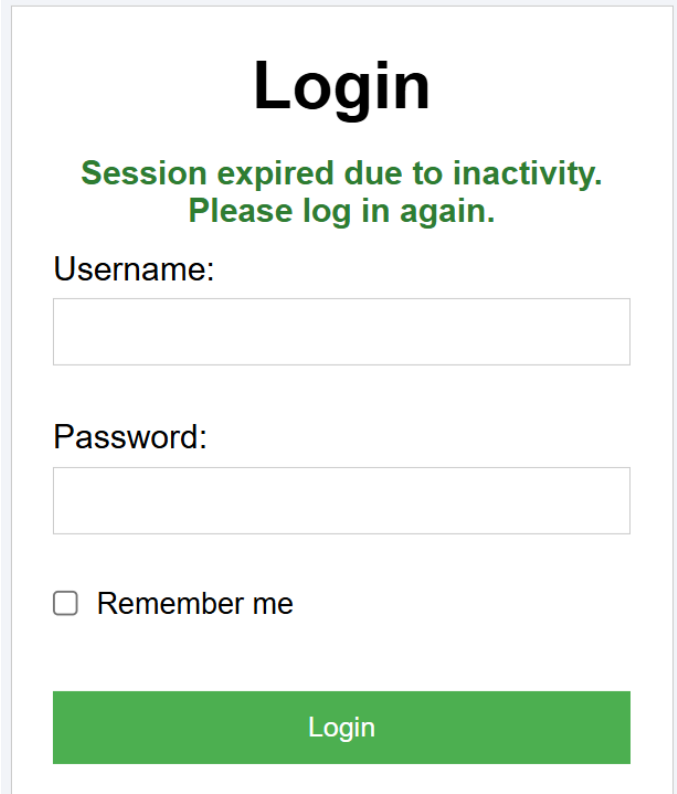
Student Login:



Student DashBoard:



Session timeout(after refreshing page):



5.Github Link:

https://github.com/HimeshMale/Student-Hub-Portal.git

6. Challenges Faced:

I wasn't able differentiate admin and student dashboard like how should I differentiate both.

7. Learning Outcomes:

I learned how to use data as creadentials at webpage.

Faculty Remarks: